

Quest CCS Project — Blue Hydrogen Production with Geological Storage

Five Million Tonnes Stored: Industrial CCS at Shell's Scotford Complex

QUICK FACTS

Sector: Oil Sands Upgrading / Blue Hydrogen Production

Location: Scotford, Alberta, Canada

Technology: Pre-combustion CO₂ capture from hydrogen production; deep saline aquifer injection

1. Background: Oil Sands and the Case for Industrial CCS

Alberta's oil sands produce one of the most energy-intensive forms of petroleum on Earth. Upgrading raw bitumen into synthetic crude oil requires large quantities of hydrogen — produced by Steam Methane Reforming (SMR) of natural gas, which generates substantial CO₂. Shell's Scotford Upgrader, located northeast of Edmonton, is one of the largest oil sands upgrading facilities in Canada.

The Quest CCS project was developed to capture the CO₂ produced by three SMR hydrogen plants at the Scotford complex and inject it into a deep saline aquifer. The project received CAD 865 million in government funding (CAD 745 million federal + CAD 120 million Alberta) and CAD 865 million in Shell investment — a total of approximately USD 1.35 billion.

Operator	Shell Canada Energy
Location	Scotford, Alberta, Canada (54°N, 113°W)
Commissioned	November 2015
Source of CO ₂	Three steam methane reforming (SMR) hydrogen plants
Designed capture capacity	1.08 million tonnes CO ₂ /year
Actual capture (typical year)	~1.0–1.08 Mt/year
Total stored (to 2024)	~7 million tonnes CO ₂
Capture technology	Benfield process (potassium carbonate absorption) — pre-combustion
Storage formation	Basal Cambrian Sands, 2,000 m depth
Number of injection wells	3 (with monitoring well array)

Monitored leakage	Zero — 9 years of continuous monitoring
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2. Technology: Pre-Combustion Capture from SMR

Quest captures CO₂ using a fundamentally different approach from Boundary Dam's post-combustion process. In Steam Methane Reforming, the capture opportunity occurs before combustion — at the point where CO₂ is produced as an intermediate product:

Step	Chemical Process	Product
SMR reaction	$\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$ (at ~850°C, with catalyst)	Syngas (CO + H ₂)
Water-gas shift	$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$ (at ~200–450°C)	H ₂ + concentrated CO ₂
CO ₂ removal (Quest)	Benfield process: K ₂ CO ₃ absorbs CO ₂ selectively	Pure H ₂ stream + CO ₂ stream
H ₂ use	Hydrogen used in upgrading process (desulphurisation, hydrocracking)	Oil sands synthetic crude
CO ₂ compression	CO ₂ compressed to ~20 MPa supercritical state	Ready for pipeline transport
Pipeline transport	84 km pipeline to injection site	CO ₂ at injection wellhead
Geological injection	CO ₂ injected into Basal Cambrian Sands at 2,000 m	Permanently stored underground

The pre-combustion capture advantage: The CO₂ concentration in the SMR process stream after the water-gas shift reaction is approximately 15–25% — far higher than the 12–15% in coal power flue gas, and dramatically higher than the 0.04% in ambient air for DAC. Higher concentration = lower capture energy penalty = lower cost per tonne.

3. Performance: A Project That Works

Unlike Boundary Dam, which faced significant performance gaps in its early years, Quest has operated close to its design targets from commissioning onwards. This is partly because the capture technology (Benfield process) was already mature and well-understood from its use in industrial gas processing — Quest adapted existing technology rather than pioneering new processes.

Year	CO ₂ Captured (Mt)	Capture Rate	Notes
2015 (partial)	0.50	~98%	Commissioning and ramp-up
2016	1.03	~95%	First full year

Year	CO ₂ Captured (Mt)	Capture Rate	Notes
2017	0.88	~82%	Planned maintenance outage
2018	1.08	~100%	Best performance year
2019	1.02	~94%	Normal operations
2020	0.76	~70%	COVID-19 reduced production
2021	1.06	~98%	Recovery to normal
2022	1.07	~99%	Strong performance
2023–2024	~1.0/year	~93%	Steady state

4. The Storage Site: Basal Cambrian Sands

The CO₂ from Quest is injected into the Basal Cambrian Sands (BCS) — a deep saline aquifer formation that extends across much of Alberta. The formation's properties make it well-suited for CO₂ storage:

- **Depth (2,000 m):** Sufficient for CO₂ to be in supercritical state at formation temperature and pressure. No risk of CO₂ boiling or reverting to gas phase.
- **Injectivity:** The BCS has good permeability in the Scotford area, allowing injection at planned rates without excessive pressure build-up.
- **Seal integrity:** Multiple impermeable rock layers above the BCS provide redundant sealing. The thickest seal unit — the Proterozoic Shaftesbury formation — has been undisturbed for hundreds of millions of years.
- **Monitoring programme:** Three dedicated observation wells, 4D seismic, pressure monitoring, and periodic sampling of groundwater above the formation. Zero anomalies detected after 9 years.

5. Economics and Policy Implications

Quest's economics were heavily dependent on government funding — the project would not have been built at the prevailing carbon price of 2012–2015. However, its performance has been used to refine cost estimates for similar industrial CCS projects:

Cost Component	Quest (First-of-a-Kind)	Projected (Nth-of-a-Kind, 2030)
Capture capital	~\$110/tonne (annualised)	~\$60–80/tonne
Storage (pipeline + injection)	~\$25/tonne	~\$15–20/tonne
Monitoring	~\$5/tonne	~\$3–5/tonne
Total cost	~\$140/tonne	~\$80–105/tonne
Canadian 45Q equivalent value	~\$30/tonne (2015)	~\$85/tonne (2023 US 45Q)

Cost Component	Quest (First-of-a-Kind)	Projected (Nth-of-a-Kind, 2030)
Gap covered by government	~\$110/tonne	~\$0–20/tonne at \$80/t carbon price

6. The Blue Hydrogen Debate and India's Context

Quest has been central to debates about 'blue hydrogen' — hydrogen produced from natural gas with CCS. Critics argue that methane leakage from the natural gas supply chain undermines blue hydrogen's climate credentials. Defenders point to Quest's actual capture data and note that blue hydrogen with well-managed gas supply chains performs significantly better than grey hydrogen (no capture).

- **India's hydrogen strategy:** India's National Green Hydrogen Mission targets 5 million tonnes of green hydrogen by 2030. But in the near term, blue hydrogen from CCS-equipped SMR offers a faster, lower-cost pathway for sectors like fertiliser production and oil refining — India's largest current hydrogen users.
- **Indian fertiliser sector:** India's 33 major fertiliser plants produce ammonia from natural gas, generating ~60 million tonnes of CO₂ per year. CCS on Indian ammonia plants — using a similar approach to Quest — could be one of the most cost-effective industrial CCS applications in India.
- **Cost trajectory:** Quest's demonstrated cost of ~\$140/tonne will fall substantially with scale and learning. Indian SMR+CCS on fertiliser plants benefits from: (a) larger scale, (b) higher CO₂ concentration (cheaper to capture), (c) lower labour costs, and (d) proximity to potential geological storage in Indian sedimentary basins.

Key Takeaway: Quest proves that pre-combustion CCS from SMR hydrogen plants can achieve near-100% capture rates continuously over many years. For India's fertiliser sector, which accounts for a massive share of industrial CO₂, this technology pathway deserves serious attention.